Defining Research Reproducibility: What Do You Mean?

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Vague definitions and an unclear understanding of "research reproducibility" have contributed to confusion in the scientific community. In a recent article published in *Science Translational Medicine*, Goodman et al. showed that the number of studies concerning research reproducibility has been steadily increasing since the 1970s, and that reproducibility and replicability are frequently used interchangeably *(1)*. According to a US National Science Foundation subcommittee on replicability in science, reproducibility is the ability to duplicate results using the same raw data; replicability is the ability of a study to duplicate results with newly collected data *(2)*. To illustrate confusion concerning these terms, a former director of the US National Institutes of Health (Francis Collins) has used the word *reproducibility* as the ability of a study to collect new data points and obtain results corroborating another study *(3)*.

In their article, Goodman et al. introduced adjectives to help clarify the *type* of reproducibility being assessed *(1)*:

- Methods reproducibility: the ability to use the same experimental/computational procedures, data, and tools to obtain the same results. According to the US National Science Foundation subcommittee on replicability in science *(2)*, it is the "minimum necessary condition for a finding to be believable and informative."
- Results reproducibility: the ability to corroborate results using the same methods in another study.
- Inferential reproducibility: the ability to draw similar findings through the replication or reanalysis of a study.

Assessing whether a study achieves any of these definitions is more complex because different fields of study require different degrees of agreement. For example, assessing methods reproducibility in a mathematical study may require sharing basic data, software, and systems used, whereas a chemical study would require the same plus more granular information, like manufacturers, reagents, and lots used.

Results reproducibility should also be assessed by field (e.g., mathematical studies require close agreement of results, whereas more imprecision is allowed in psychology). The article argues that the use of statistical methods such as *P* values and metaanalysis *can* be deceptive when compounded by multiplicity such as publication bias. To avoid this, they recommend evaluation of the evidence either study provides to a hypothesis *(1)*.

Finally, inferential reproducibility deals with the conclusions drawn from data. Identical analysis could result in different conclusions by different researchers, whereas the same conclusions could be drawn from different studies with divergent data and analysis.

Because of these complexities, the authors recommend using a more cumulative approach to knowledge acquisition. A Bayesian system allows for the pretest probability to be weighed with the strength of results from the experiment. For example, an unlikely hypothesis requires large quantities of good evidence before it is assessed as true. To conclude, Goodman et al. postulate that poor reproducibility and redundant research could be reduced if scientists strive to better characterize the type of reproducibility being assessed, have a deeper understanding of the implications of reproducibility on specific hypotheses, and have a clearer understanding of the limitations of statistical significance *(1)*.

Author Contributions: *All authors confirmed they have contributed to the intellectual content of this paper and have met the following 3 requirements: (a) significant contributions to the conception and design, acquisition of data, or analysis and interpretation of data; (b) drafting or revising the article for intellectual content; and (c) final approval of the published article.*

Authors' Disclosures or Potential Conflicts of Interest: *No authors declared any potential conflicts of interest.*

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Received July 21, 2017; accepted July 26, 2017.
DOI: 10.1373/clinchem.2017.279984
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